

Commercial Auto Risk Scoring Case Study

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Table of contents

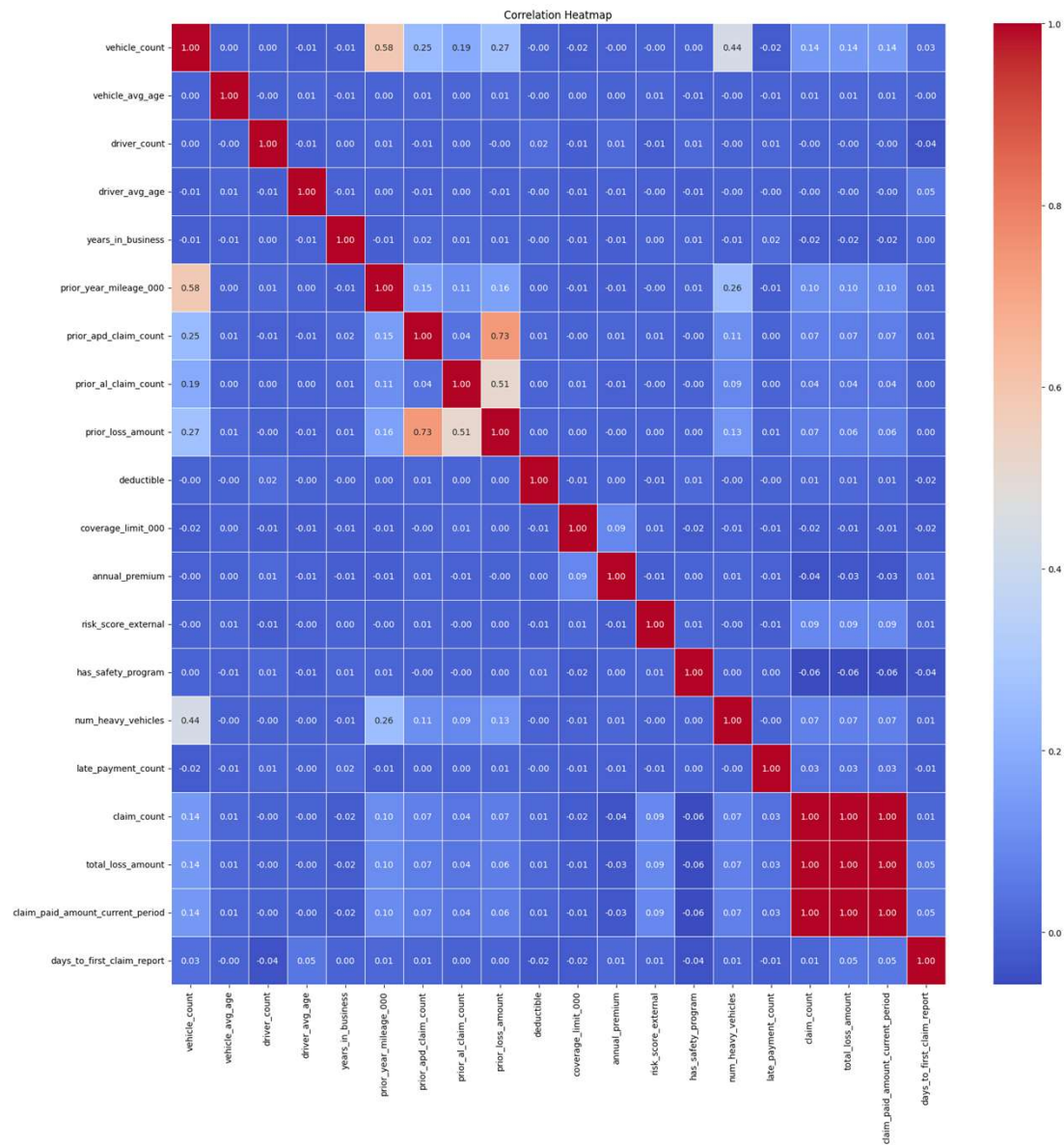
- Exploratory data analysis
- Modeling methodology improvements
- Production readiness improvements
- Deployment roadmap

Explanatory data analysis (EDA)

- We have plenty null values in different columns
- We are working with data types:
 - numeric – int, float
 - object – categorical data, dates
- Columns that are not present in both dataframes:
 - labels – claim_count, total_loss_ammount
 - other – has_safety_program
- Duplicate rows:
 - full duplicates – 42 records
 - duplicate id (policy_id) – 33 records

Explanatory data analysis (EDA)

- Attributes are not normally distributed – data is skewed:
 - normaltest, histogram
- Most numeric attributes have outlier values
- Relationships and correlations:
 - Used spearman correlation
 - Data leaks: Claim_paid_ammount_current_period – has value only when labels contain value > 0
 - Correlation $\approx$ 100%
 - Multicollinearity:
 - vehicle_count - prior_year_mileage_000
 - prior_apd_claim_claim_count - prior_loss_amount
 - prior_al_claim_count - prior_loss_amount



Modeling methodology improvements

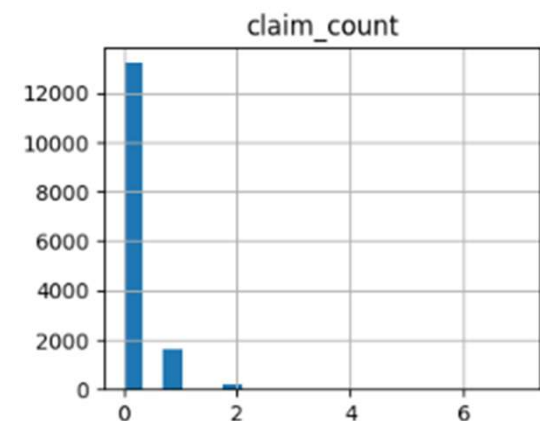
Different feature selection

- Current split used -> `X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)`
- This split is created with unordered randomly shuffled data
- More fitting would be splitting after ordering based on year – time-based
- Results of time-based validation set would be more generalizing
- Years (or months) can be extracted from `policy_effective_date`

Modeling methodology improvements

Imbalanced target column

- EDA showed us how imbalanced our label attribute – claim_count
- Vast majority of records are without claim (13146)
- If predictions are all 0, we still get great accuracy
- Solution: `tmp = LogisticRegression(max_iter=1000, class_weight='balanced')`



Modeling methodology improvements

Different feature selection

- Model is fed leaky columns:
 - `claim_paid_ammount_current_period` – shows 100% correlation with our label columns
 - `first_claim_reported_date`, `claim_status_current_period`, `days_to_first_claim_report`:
 - Found by manual observation
 - If they do not contain specific values (null/ No Claim) they predict target values
- Model is fed multicollinear attributes
 - Filter them with cross validation

Production readiness improvements

- Using pipeline:
 - Provides end-to-end execution
 - Eliminates training-serving-skew
 - Replaces current manual preprocessing
- Modular code architecture:
 - Separating one script to several modules
 - Improves readability
 - Improves scalability
- Adding schemas for data validation:
 - Automatic solution to detect possible threats
 - Deals with corrupt values
 - Deals with missing values/ columns
 - Ensures Model is fed safe records

Deployment Roadmap

Containerization

- Package code, model, pipeline into a Docker container
- Ensures environment compatibility

Testing

- Creating scenarios for edge cases
- Performance and load testing
- Regression testing

Deployment Roadmap

Monitoring

- Track prediction quality, compare obtained results
- Setting alerts, triggers in case of errors, or quality drop

Documentation

- Record data sources
- Track versions
- Technical and operational documentation

AI use

- **Models used**

- Gemini by Google (3.1 pro/ 3.5 Flash-lite)

- **Data navigation**

- Used to parse, navigate, and understand unfamiliar insurance dataframes

- **Methodological correction**

- Challenged initial assumptions and identified tailored methods for specific scenario of the given task

- **Research**

- Used to research complex topics, such as specifics of deployment roadmap

Thank you for your attention!